

1. Graph the following piecewise function: $f(x) = \begin{cases} x & \text{if } x \geq 2 \\ x^2 - 3 & \text{if } 0 \leq x < 2 \\ 5 & \text{if } x < 0 \end{cases}$

2. In your graph from #1, change the linear pieces of the function so that they have the same starting points, but are now quadratic. Create a new possible formula for this function.

3. Graph the following piecewise function: $f(x) = \begin{cases} 3x - 5 & \text{if } x > 4 \\ \ln(x) & \text{if } 0 < x < 4 \end{cases}$

4. Graph the following adaptation of the function from #2: $f(x) = \begin{cases} 3x - 5 & \text{if } x > 4 \\ \ln(x) & \text{if } 0 < x < 4 \\ e^x & \text{if } x \leq 0 \end{cases}$